

Statement of intent

Baha Khammari · PhD position, auction-based support schemes for renewable energy
LIDAM Energy Group, CORE, UCLouvain

I want to write a doctorate on how the design of renewable support schemes changes what gets built, and what I would bring is identification. Energy market design is argued mostly in theory, while Europe has spent a decade running the experiments. Member states adopted auctions at different dates, under different rules, and revised them on their own timetables. Reading that variation is work for an applied microeconomist. Both directions I would take up rest on one observation from my own research: the support an instrument appears to offer and the incentive it creates are different quantities, and the rules decide the distance between them.

My training is in economics at the University of Cologne, where the energy specialisation is two modules. *Growth, Energy, Climate Change* covers the physical and economic foundations of energy systems and European regulation and market intervention. *Energy Markets and Regulation*, with Jun.-Prof. Dr. Oliver Ruhnau, takes those into grids and interconnectors and the European electricity market. I completed it with 1.3, where 1.0 is the maximum. Those modules are why I read an auction rule as a physical and regulatory constraint and not only an economic one. My quantitative preparation is the strongest part of my record: mathematics for economists 1.0, econometrics 1.3, and 1.0, 1.0 and 1.3 in bachelor statistics, mathematics and applied econometrics. I have also taught it, tutoring two school students through Abitur-level mathematics, which is the test of whether you know a method or only recognise it. My thesis, graded 1.0, put that to work on the labour-market effects of a national scholarship programme in Brazil, on a panel I built from administrative records, five million linked employer-employee observations across 558 microregions, 2005 to 2019, in R through BigQuery. Callaway and Sant'Anna (2021) with doubly-robust inference gives a wage gain of 3.04 per cent in low-exposure regions, distinguishable from zero at the 10 per cent level, on a non-monotonic dose-response. What that settled is that I can take a staggered policy rollout from raw records to a defensible causal estimate on my own.

The two directions I would take up are de-risking green investment and multi-technology auction design. My final seminar paper already sits inside the first. It defines the effective EU ETS carbon cost as the allowance price attenuated by the sector-year free-allocation share, and shows the attenuation is not lump-sum, because activity thresholds and closure provisions tie allocation to continued production. A public intervention that weakens a cost incentive while the headline price is unchanged is the mechanism your first direction names, and the same logic runs through guarantees, CfDs and public insurance. The second direction has the cleaner empirical footing. Since the 2016 state aid framework member states have arrived at technology-neutral, technology-specific and mixed designs at different dates, and Germany's joint wind and solar auction of April 2018 gave the entire 200 MW to solar at bids above the technology-specific benchmark. Portfolio composition and rents are observable, which makes the question measurable.

Three people at CORE decide where I want to do it. Your work with Yueting Yu on discriminatory against uniform-price auctions asks how short-run bidding maps into long-run investment, the rent question inside the second direction. Quentin Lété works on market design, capacity investment and risk management, which is the first. Enerpol, with Jean Hindriks and François Maniquet, studies who carries the cost of the transition, which is what my thesis estimates. In ten years I want to be running the empirical side of European energy market design, publishing the evidence that auction and support-scheme reform gets judged on. Reaching it needs two things I do not have yet, graduate training in the theory this group works in and four years of supervised research on a question I already care about. This position is both. I would be glad to discuss it with you.

Available from October 2026. French B1, German and Tunisian Arabic native, English IELTS Academic 8.0. EU citizen. Referees: Prof. Dr. Pia Pinger and Jun.-Prof. Dr. Oliver Ruhnau, contact details on the CV.

Baha Khammari · b.e.khammari@gmail.com · +49 1520 9016956